Portfolio RiskQ

Classical-to-Quantum Tail Risk Estimation

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Introduction

Portfolio risk requires accurate tail estimates such as VaR and CVaR, yet production systems must balance accuracy with compute cost. Portfolio RiskQ explores a classical-to-quantum workflow that preserves interpretability while opening a path to faster tail estimation.

Condensed Abstract

We built an end-to-end pipeline to estimate portfolio tail risk, spanning data calibration, stress scenarios, nested Monte Carlo, discretization into 2^n bins, and an IQAE simulator that estimates tail probability with confidence intervals.

Background

VaR and CVaR summarize the loss tail that matters most to risk teams, while nested Monte Carlo provides a robust classical baseline. Quantum amplitude estimation promises a quadratic reduction in sampling complexity, but it requires discretized, quantum-ready distributions. Think of VaR as a "loss line you rarely cross," and CVaR as the "average loss when you do cross it." In plain language, we simulate many possible futures, summarize the worst losses, and compare a classical estimate with a quantum-inspired estimate of the same tail risk. We treat returns as random and generate many loss outcomes for the portfolio. Intuition: the quantum step is a smarter way to estimate a probability (the tail) with fewer effective samples. Reading the example: the tail probability is about 0.83%, and the interval shows expected uncertainty. This makes the classical vs quantum comparison fair and easy to explain to non-specialists.

Hypothesis

IQAE-style estimation on discretized loss distributions can approximate tail probability with fewer effective samples while preserving interpretability for risk stakeholders.

Objectives

- Build a working classical risk engine and dashboard MVP.
- Add discretization and tail-mask generation.
- Deliver a small-scale IQAE simulator with resource reporting.

The emphasis is on a complete, inspectable pipeline that supports both classical validation and quantum-oriented estimation.

Research

- Literature review of VaR/CVaR and nested Monte Carlo.
- IQAE fundamentals and Grover iteration behavior.
- Practical constraints for near-term quantum workflows.

This framing informs where quantum methods can add value without compromising auditability.

Models

- Gaussian return model; optional crash-mixture scenario.
- Nested Monte Carlo for regime uncertainty.
- Histogram-based discretization for quantum mapping.
- Backend toggle: classical vs quantum comparison path.

Materials

Implementation uses Python with NumPy and Pandas for numerical work, a FastAPI backend for serving risk metrics, and a Streamlit + Plotly front end for interactive visualization.

Methodology

Nested Monte Carlo (loss samples)

$$r \sim \mathcal{N}(\mu, \Sigma), \quad L = -w^\top r$$

Tail metrics

$$\mathrm{VaR}_\alpha = Q_\alpha(L)$$
$$\mathrm{CVaR}_\alpha = \mathbb{E}[L \mid L \geq \mathrm{VaR}_\alpha]$$

Discretization

- Bin losses into 2^n intervals.
- Produce edges, probabilities, tail mask.
- Bootstrap error estimate for bin stability.

IQAE (simulator)

$$p = \sin^2(\theta), \quad p_m = \sin^2((2m+1)\theta)$$

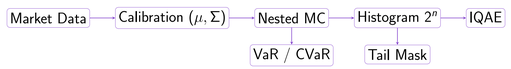
- Uses Grover iteration schedule to amplify tail states.
- Returns estimate with confidence interval.

Analysis

- Validate probability normalization and tail-mask correctness.
- Compare IQAE estimates to true tail probability on toy data.
- Track resource usage (qubits, depth, 2Q gates, oracle calls).
- Compare classical vs quantum using same ℓ and histogram.

All comparisons share the same tail threshold ℓ, scenario configuration, and histogram to ensure a fair baseline.

System Pipeline (Schematic)



Results

- Discretization tests pass (prob sums to 1, mask correctness).
- IQAE toy test estimates match true tail probability.
- Dashboard shows VaR/CVaR and tail-highlighted histogram.
- Backend toggle: classical vs quantum with side-by-side comparison.
- Difference fields (abs/rel error) reported in API/ UI.

Together these results show that the quantum comparison can be presented alongside classical metrics without changing the underlying scenario or data pipeline.

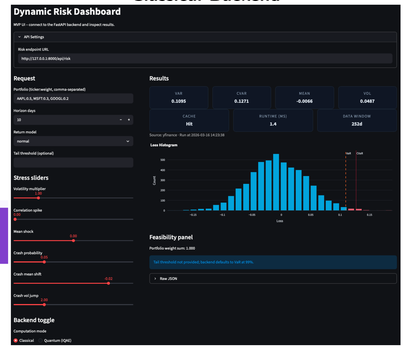
Results Snapshot

- Tail probability (classical) vs IQAE estimate with CI.
- Histogram padding to power-of-two bins when required.
- Runtime and cache indicators for reproducibility.
- Example output: $\hat{p} = 0.0083$ with CI $(0.0079, 0.0087)$.

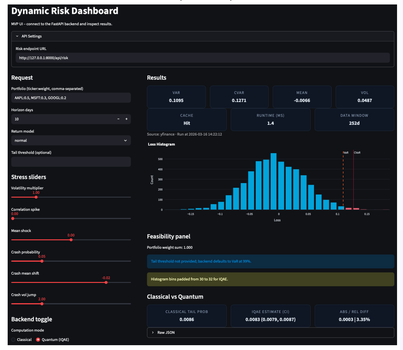
Dashboard Visuals

Left panel collects portfolio inputs and stress settings; right panel reports VaR/CVaR, histogram with tail highlight, and a classical vs quantum comparison when enabled. The classical mode reports tail probability from the histogram directly, while quantum (IQAE) estimates the same tail probability with a confidence interval and reports absolute/relative error.

Classical Backend



Quantum (IQAE) Backend



Discussion

- Discretization provides a stable bridge to IQAE.
- Simulator results align with expected tail probabilities.
- Stress sliders need full backend integration.
- Comparison uses same threshold ℓ, scenario, and histogram.
- Padding can affect tail resolution at small bin counts.

Padding effects are explicitly tracked to avoid overstating precision at small bin counts.

Implications

- Feasible path to hybrid classical-quantum risk pipelines.
- Resource reports inform near-term quantum limits.
- Clear audit trail from inputs to tail estimate.

This enables clearer communication of quantum feasibility to non-quantum stakeholders.

Conclusion

Portfolio RiskQ demonstrates an end-to-end workflow from classical risk estimation to quantum-ready tail probability estimation using IQAE, with consistent inputs and transparent reporting across both modes.

Conclusion

The current MVP validates feasibility of a classical-to-quantum pipeline and establishes a foundation for future quantum execution.

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